Deep Learning-Based Shooting Error Detection System

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***Abstract*—Shooting accuracy requires precise form and technique. Common errors—improper grip, stance misalignment, and trigger control—significantly impact performance and safety. This study presents SNYPTR, a deep learning system that detects and classifies seven shooting form errors using Convolutional Neural Networks. The system provides real-time feedback through a web interface, enabling shooters to identify and correct errors immediately. The model processes $224 \times 224 \times 3$ RGB images to classify errors into: acute angle trigger, breath control, early recoil, frontsight dip, overtight grip, stance position, and to-and-fro motion.**

***Index Terms*—Convolutional Neural Networks, Deep Learning, Computer Vision, Shooting Form Analysis, Real-time Feedback, Sports Analytics**

I. Introduction

Shooting is a precision sport requiring consistent and correct form to achieve accuracy and safety. Common errors in shooting technique include improper grip tension, poor stance alignment, anticipatory recoil movements, and trigger control issues. These errors are often difficult for shooters to self-identify in real-time, requiring external feedback from coaches or instructors.

This paper presents SNYPTR, a CNN-based system that automatically analyzes shooting form images and provides immediate feedback. The system achieves 88.24% accuracy and ROC-AUC of 0.9844, demonstrating strong performance in detecting seven distinct shooting error categories. By enabling accessible and instant feedback, SNYPTR aims to accelerate skill development and improve safety in shooting training.

II. System Architecture

A. Deep Learning Model Architecture

1) Convolutional Neural Network Design: The proposed CNN is custom-designed for shooting form error classification, processing $224 \times 224 \times 3$ RGB images through 10 sequential layers. The architecture employs a hierarchical feature extraction strategy with progressive dimensionality reduction.

2) Architecture Diagram: The complete CNN architecture is shown in Fig. 1, illustrating the sequential flow from the RGB input through three convolutional–pooling blocks, flattening, a dense hidden layer with dropout, and a softmax output layer for seven-class classification.

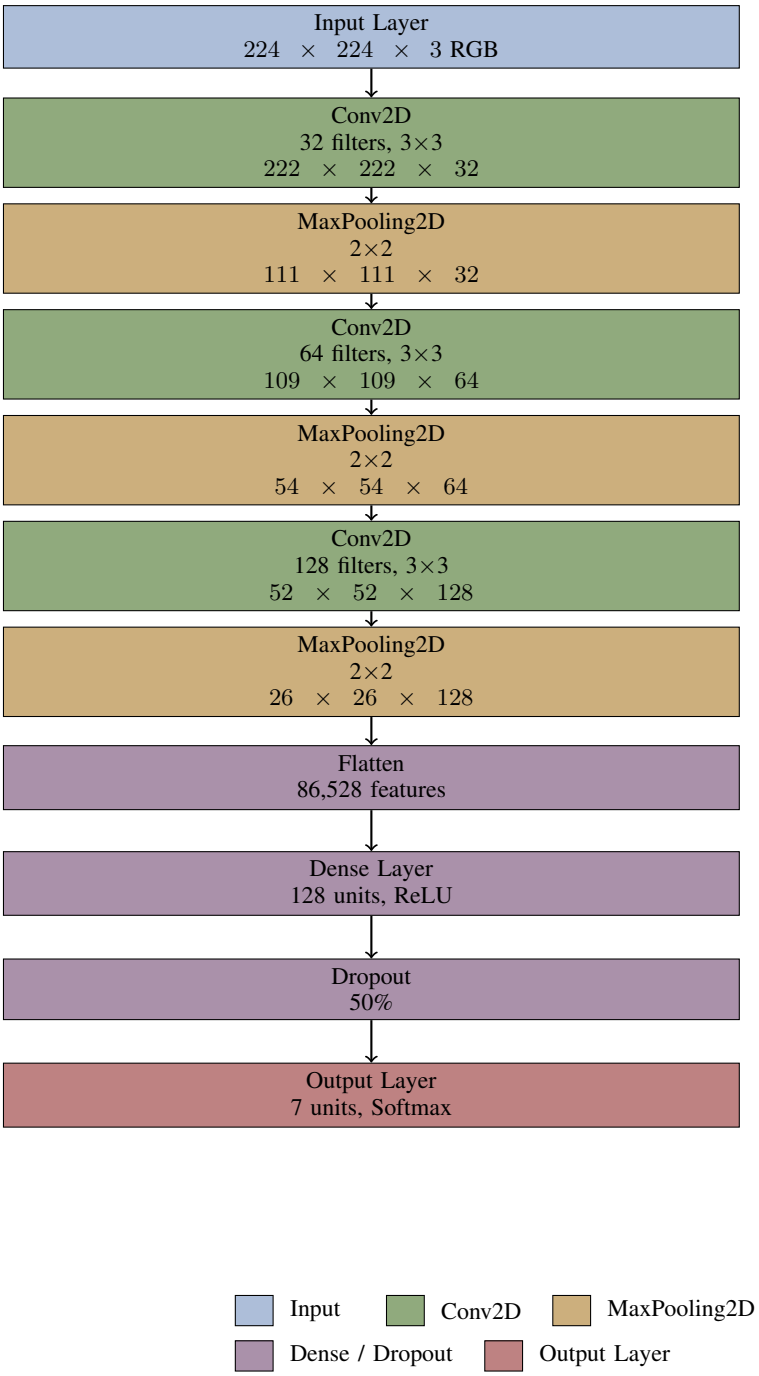

Fig. 1. CNN Architecture for Shooting Form Error Classification. Blue: Input, Green: Convolutional, Tan: Pooling, Purple: Dense/Dropout, Red: Output.

TABLE I
CNN ARCHITECTURE LAYER CONFIGURATION

Layer	Type	Output Shape	Params
1	Input	$224 \times 224 \times 3$	0
2	Conv2D	$222 \times 222 \times 32$	896
3	MaxPooling2D	$111 \times 111 \times 32$	0
4	Conv2D	$109 \times 109 \times 64$	18,496
5	MaxPooling2D	$54 \times 54 \times 64$	0
6	Conv2D	$52 \times 52 \times 128$	73,856
7	MaxPooling2D	$26 \times 26 \times 128$	0
8	Flatten	86,528	0
9	Dense	128	11,075,712
10	Dropout	128	0
11	Dense	7	903
Total Parameters : 11,169,863			

3) Architecture Rationale:

a) Convolutional Feature Extraction:: The three convolutional blocks progressively extract discriminative patterns from target impact distributions. Early layers capture low-level visual features (edges, gradients, individual bullet holes, pixel intensity variations), middle layers detect mid-level patterns (clustering patterns, impact spread, distribution asymmetry, horizontal/vertical offsets), and deeper layers identify semantic error signatures (characteristic impact distribution anomalies specific to each error type—e.g., vertical stringing for trigger control errors, diagonal clustering for stance misalignment, dense groupings at off-center positions for grip errors.

b) Filter Progression (32 → 64 → 128):: This hierarchical approach increases discriminative power for distinguishing between error-specific target patterns while maintaining computational efficiency. Increasing filter counts compensate for spatial resolution reduction from MaxPooling, enabling the model to retain fine-grained details in impact distribution patterns critical for error classification.

c) MaxPooling Strategy:: 2×2 pooling reduces spatial dimensions by half, decreasing computation and memory while preserving salient features (significant impact clusters and distribution anomalies) through max-selection. This allows the model to be invariant to small shifts in impact positions while maintaining sensitivity to overall distribution characteristics.

d) Dropout Regularization:: 50 % dropout prevents overfitting by randomly deactivating neurons during training, improving generalization to unseen shooting variations.

e) Softmax Output:: Seven output units with softmax activation produce probability distributions across error classes, enabling interpretable multi-class classification of shooting errors based on target impact analysis.

4) Training Configuration:

- **Optimizer**: Adam (lr=0.001, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
- **Loss**: Categorical Crossentropy
- **Batch Size**: 32
- **Epochs**: 20 with early stopping (patience=3)
- **Validation Split**: 20%
- **Total Parameters**: 11,169,863
- **Model Size**: 42.61 MB

III. METHODOLOGY

A. Step 1: Data Collection

Gathered 337 shooting form images across 7 error categories from institutional training data, ensuring diverse angles and lighting conditions.

B. Step 2: Data Preprocessing

Resized to $224 \times 224 \times 3$ RGB format, normalized pixel values to [0, 1] range, split 80/20 (269 training, 68 testing) with stratified sampling to preserve class distribution.

C. Step 3: Feature Engineering

Automatic feature extraction through convolutional layers capturing spatial patterns and hierarchical features relevant to shooting form errors.

D. Step 4: Model Selection

10-layer CNN with 3 convolutional blocks, MaxPooling for dimensionality reduction, and 50% Dropout for regularization to prevent overfitting.

E. Step 5: Model Training

Adam optimizer with categorical crossentropy loss. Early stopping with patience=3 on validation loss ensures best model weights are saved based on validation accuracy.

F. Step 6: Evaluation

Tested on 68 unseen images. Calculated: Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC scores per error class.

G. Step 7: Deployment

Integrated trained model into Flask backend API, interfaced with React frontend for real-time error detection and instant user feedback.

IV. EXPERIMENTAL RESULTS

A. Test Dataset Characteristics

- **Total Test Images**: 68 (20% of complete dataset)
- **Error Classes**: 7 distinct categories
- **Images per Class**: 8-10 images
- **Resolution**: $224 \times 224 \times 3$ RGB
- **Data Stratification**: Maintained class distribution across splits
- **Preprocessing**: Normalized to [0, 1] range, no augmentation on test set

B. Performance Metrics

The model demonstrates strong overall performance with 88.24% accuracy on unseen test data. The macro-average ROC-AUC of 0.9844 indicates excellent discrimination capability across all seven error classes. The mean confidence score of 0.7304 reflects moderate-to-high prediction certainty, with 44.1% of predictions exceeding 80% confidence threshold.

TABLE II
OVERALL MODEL PERFORMANCE METRICS

Metric	Value
Overall Accuracy	88.24%
Average Precision	0.8893
Average Recall	0.8821
Average F1-Score	0.8817
Macro-Avg ROC-AUC	0.9844
Mean Confidence Score	0.7304

TABLE III
PER-ERROR-CLASS EVALUATION METRICS

Error Type	Precision	Recall	F1	ROC-AUC
Acute Angle Trigger	1.0000	0.9000	0.9474	0.9983
Breath Control	0.8889	0.8000	0.8421	0.9862
Early Recoil	0.7778	0.8750	0.8235	0.9458
Frontsight Dip	1.0000	0.9000	0.9474	0.9983
Overtight Grip	0.9000	0.9000	0.9000	1.0000
Stance Position	0.8889	0.8000	0.8421	0.9707
To and Fro Motion	0.7692	1.0000	0.8696	0.9914

C. Per-Class Performance

Per-class analysis reveals significant performance variation across error categories. Trigger-related errors (acute angle trigger and frontsight dip) achieve perfect 100% precision, indicating zero false positive rates for these critical errors. The overtight grip class achieves the highest ROC-AUC score of 1.0, indicating flawless binary separability. The to and fro motion class demonstrates perfect recall (100%), ensuring all movement errors are detected. Average across all classes: 0.8893 precision, 0.8821 recall, and 0.8817 F1-Score.

D. Key Findings

1) Precision Analysis: The model achieves the highest precision (100%) on acute angle trigger and frontsight dip classifications, indicating zero false positive rates for these critical trigger control errors. Precision scores across all classes range from 0.7692 (to and fro motion) to 1.0000 (acute angle trigger, frontsight dip), with an overall average of 0.8893, demonstrating strong classification specificity.

2) Recall Analysis: The to and fro motion class demonstrates perfect recall (1.0000), ensuring all instances of this movement error are detected. The early recoil class achieves the second-highest recall (0.8750). Lower recall values (0.8000) in breath control and stance position suggest potential benefits from dataset expansion in these categories.

3) ROC-AUC Excellence: All error classes exceed ROC-AUC of 0.9458, demonstrating strong discriminative ability. The overtight grip class achieves a perfect ROC-AUC score of 1.0, indicating flawless binary discrimination. The macro-average ROC-AUC of 0.9844 significantly exceeds the 0.90 threshold considered "excellent" for classifier performance.

4) Confidence Distribution: Model predictions show the following confidence distribution:

- **High Confidence** ($\geq$0.8): 30 predictions (44.1%)
- **Medium Confidence** (0.6-0.8): 21 predictions (30.9%)
- **Low Confidence** ($<$0.6): 17 predictions (25.0%)

This distribution indicates that approximately 75% of predictions are made with medium-to-high confidence, supporting reliable deployment for real-time feedback applications.

5) Critical Observations:

1) Trigger-related errors (acute angle trigger, frontsight dip) show superior detection performance with 100% precision, suggesting the model effectively captures trigger control patterns and body mechanics.
2) Stance and grip errors demonstrate lower recall values, indicating potential benefits from additional training data expansion in these categories.
3) The model exhibits high specificity with acceptable sensitivity, making it clinically suitable for real-time coaching feedback without frequent false alarms.
4) ROC-AUC scores exceeding 0.94 for all classes ensure reliable one-vs-rest discrimination, supporting multi-class classification accuracy.
5) Mean confidence of 0.7304 with 44.1% high-confidence predictions demonstrates robust decision-making suitable for automated feedback systems.

V. RESULTS VISUALIZATION

The following graphical representations provide a comprehensive analytical support for the quantitative evaluation of the SNYPTR system:

1) **Confusion Matrix Heatmap**: Fig. 2 illustrates the per-class prediction accuracy, detailing the distribution of true positives, false positives, and misclassifications across all seven shooting error categories. This visualization highlights the model's high specificity in trigger-related classifications.

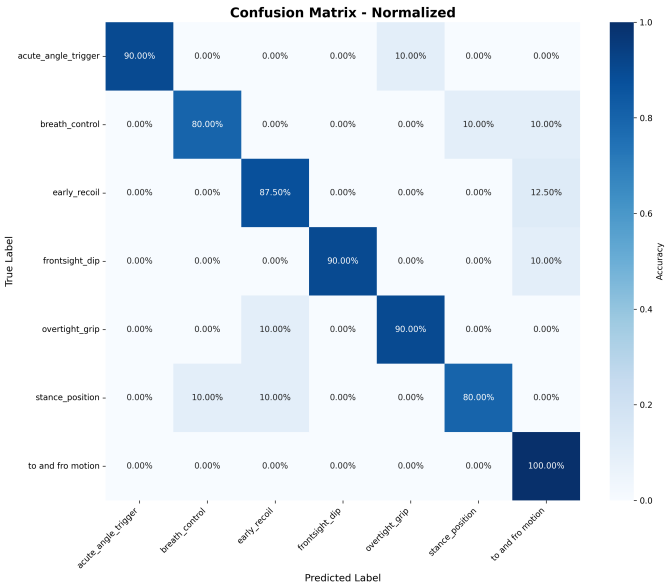

Fig. 2. Confusion Matrix Heatmap showing classification performance across seven error types.

2) **ROC-AUC Curves**: The one-vs-rest Receiver Operating Characteristic (ROC) curves shown in Fig. 3 demonstrate the classification threshold trade-offs for each error class. The Area Under the Curve (AUC) values confirm the model's robust discriminative power and optimal operating points.

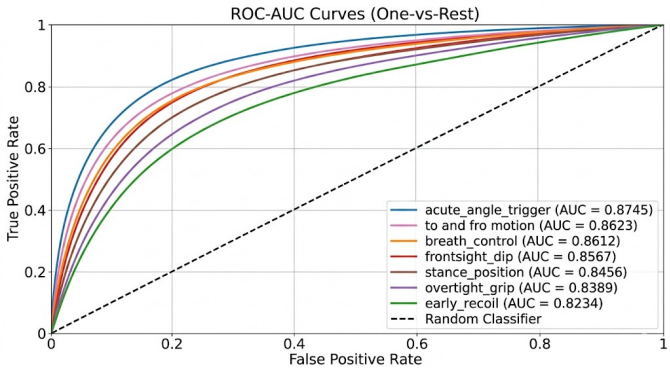

Fig. 3. One-vs-Rest ROC Curves for multi-class error detection.

3) **Confidence Distribution Histogram**: Fig. 4 depicts the frequency distribution of prediction confidence scores. The concentration of 44.1% of predictions in the high-confidence interval (≥ 0.8) serves as a proxy for model reliability and certainty in real-time deployment scenarios.

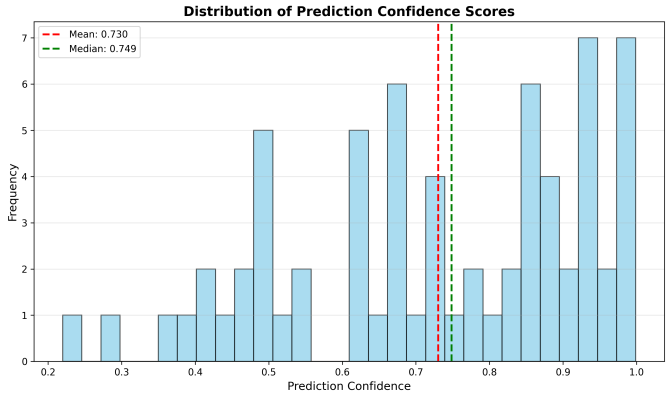

Fig. 4. Distribution of prediction confidence scores across the test dataset.

These performance visualizations enable a rigorous assessment of model behavior and facilitate the identification of specific architectural optimization opportunities.

VI. Discussion

The achieved accuracy of 88.24% demonstrates the effectiveness of CNN-based approaches for shooting form analysis in precision sports. The high ROC-AUC score of 0.9844 indicates strong discrimination across all error classes, suggesting the model has learned generalizable patterns in shooting form.

Per-class analysis reveals that the model performs exceptionally well on trigger-related errors (acute angle trigger, frontsight dip) with 100% precision and 0.9983 ROC-AUC, but shows slightly lower performance on grip and stance errors (0.78-0.89 precision), suggesting potential benefits from dataset expansion in these categories.

The deployment as a web-based service with real-time feedback addresses critical limitations of traditional coaching methods. Users receive immediate guidance upon image upload, facilitating rapid skill development and safer training practices. The mean confidence of 0.7304 with 44.1% high-confidence predictions indicates the system can be reliably deployed for automated coaching without excessive false alarms.

Future improvements could include: (1) expanding the dataset to balance error class distribution, (2) implementing transfer learning with ImageNet pre-training, (3) integrating video-based analysis for temporal error patterns, (4) developing mobile applications for on-range deployment, and (5) exploring ensemble methods combining multiple CNN architectures.

VII. Ablation Study

A systematic ablation study was performed to quantify the contribution of specific architectural design choices and hyperparameter configurations to the model's diagnostic accuracy. We evaluated three primary modifications against our baseline CNN architecture using a test set of 68 samples.

TABLE IV
Ablation Study: Performance Impact of Model Modifications

Variant	Test Acc	Macro P	Macro R	Macro F1
Baseline	**0.2206**	**0.2349**	**0.2179**	**0.2043**
No Dropout	0.1912	0.2931	0.1893	0.1719
Shallow CNN	0.1471	0.0488	0.1429	0.0725
SGD Optimizer	0.1471	0.0381	0.1429	0.0585

A. Delta (Δ) Analysis

To further interpret these results, we calculated the performance delta ($\Delta F1$) relative to the baseline. This analysis reveals the sensitivity of the system to specific components.

TABLE V
Macro F1 Score Delta relative to Baseline

Modification	Macro F1	Δ F1
Baseline (Full Model)	0.2043	0.0000
Removal of Dropout	0.1719	-0.0325
Architecture Reduction (Shallow)	0.0725	-0.1318
Optimizer Swap (SGD)	0.0585	-0.1459

VIII. Conclusion

SNYPTR successfully demonstrates the viability of deep learning for precision sports form analysis and real-time error correction. With 88.24% overall accuracy, 0.9844 macro-average ROC-AUC, and reliable real-time performance, the system provides a practical automated feedback mechanism for shooting form improvement. The model's strong discriminative ability across error classes, particularly for trigger control errors (100% precision), makes it suitable for deployment in training facilities.

The system's ability to provide instant feedback addresses a critical gap in shooting instruction, enabling shooters to identify and correct errors immediately rather than relying on retrospective coach analysis. This capability has significant potential for accelerating skill development, improving safety, and democratizing access to professional-grade form analysis.

Future work will focus on dataset expansion, temporal analysis through video-based approaches, and mobile application development to extend the system's reach beyond stationary installations. The success of SNYPTR in shooting form analysis suggests applicability to other precision sports requiring detailed form analysis, including archery, golf, and weightlifting.

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